

SECTION 15950-A
AIR AND WATER SYSTEMS TESTING, ADJUSTING, AND
BALANCING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. Section 15240, Mechanical Vibration Control.

1.2 SUMMARY

- A. Total system balance, as defined by NEBB or AABC, which constitutes the process of testing, adjusting, and balancing each system component so that the entire system produces the results for which it was designed.
- B. Work shall be performed by an independent Test and Balance Contractor retained by the Construction Manager.
- C. Work shall include the proper use of instruments, evaluation of readings, adjusting the systems to design conditions, full implementation of all test report forms, and providing unbiased opinions of the deficiencies encountered and proposing corrective action.

1.3 SCOPE

- A. The scope of work specified in this Section applies to all work required for testing, adjusting and balancing the air conditioning, heating and ventilating systems as specified herein.
- B. The work under this section shall include furnishing all labor, materials, and equipment necessary for testing, adjusting and balancing the air conditioning, heating, ventilating and exhaust systems. The work includes, but is not necessarily limited to the following:
 - 1. Testing, adjustment and balancing of air systems.
 - 2. Testing, adjustment and balancing of hydronic (water) systems.
 - 3. Measurement of final operating condition of HVAC systems.
 - 4. Sound measurement of air outlets, terminal units and equipment operating conditions.
 - 5. Rotating equipment vibration tests

- C. As part of the Work of this Contract, the equipment installation Contractor shall make any changes in the pulleys, belts and dampers or the addition of dampers required for corrected balance as recommended by Test and Balance Contractor, at no additional cost to the Construction Manager. The equipment installation Contractor shall furnish the necessary labor to assist the Test and Balance Contractor to complete its work.
- D. Test and Balance Contractor, in coordination with the equipment installation Contractors, shall visit the Project Site during construction to insure that all devices necessary to accomplish balancing have been installed.

1.4 REFERENCES

- A. The following references and standards are applicable to the testing, adjusting, and balancing of mechanical equipment and systems and shall form a part of this Section. Publication references are to the latest issue of each, together with related additions and/or amendments pertaining thereto, as of the date of contract or purchase order. Sponsoring agency references are made in accordance with the following abbreviations:
 - 1. AABC: Associated Air Balance Council
 - 2. ADC: Air Diffusion Council
 - 3. AMCA: Air Moving and Conditioning Association
 - 4. ANSI: American National Standard Institute
 - 5. ASHRAE: American Society of Heating, Refrigerating and Air-Conditioning Engineers
 - 6. ASNT: American Society for Nondestructive Testing
 - 7. NEBB: National Environmental Balancing Bureau
- B. "Procedural Standards for Testing - Balancing - Adjusting of Environmental Systems"; published by NEBB.
- C. AMCA Publication 203, "A Guide to the Measurement of Fan System Performance in the Field."
- D. ASHRAE Systems and Applications Handbook, Chapters 52 and 57 as applicable.
- E. ADC Test Code No. 1062R3, "Equipment Test Code."
- F. ANSI A 1.4, Specification for Sound Level Meters.
- G. ANSI S 1.11, Specification for Octave, Half-Octave, and Third-Octave Band Filter Sets.
- H. "National Standards for Testing and Balancing Heating, Ventilating and Air Conditioning Systems," published by AABC.

1.5 DEFINITIONS

- A. Certification of Personnel is the action of determining, verifying, or attesting in writing to the qualifications of an individual.
- B. Qualifications are the characteristics or abilities gained through training, experience, or both, that enables an individual to perform his work properly.
- C. Qualified Procedures are the test procedures developed and published by AABC, AMCA, ASHRAE, NEBB, or other nationally recognized testing, adjusting and balancing (TAB) Association or Agency.
- D. Supervision means that tests performed under this contract in any of TAB categories stated herein shall be under the direction and supervision of a TAB Level III employee on the job site.

1.6 SUBMITTALS

- A. Submit test reports in accordance with Division 1.
- B. The Test and Balance Contractor shall perform equipment and systems tests, compile test data, and submit reports to the Engineer. Specific procedures used in tests shall be performed in strict accordance with NEBB requirements and shall be included in the test report. Test and Balance Contractor shall identify equipment by Drawing identification code.
- C. Submit data on printed report forms published by AABC or NEBB.
- D. Reports shall be certified by a TAB Level III Engineer. Each individual reporting form submitted must bear the signature and TAB Level of the data recording Engineer.

1.7 REPORT FORMS

- A. Submit reports on AABC National Standards for Total System Balance forms. Submit data in SI or metric units.
- B. Forms shall include the following information:
 - 1. Title Page:
 - a. Company name.
 - b. Company address.
 - c. Company telephone number.
 - d. Project name.
 - e. Project location.
 - f. Project Engineer.
 - g. Project engineer.

- h. Project contractor.
- i. Project altitude.
- 2. Instrument List:
 - a. Instrument.
 - b. Manufacturer.
 - c. Model.
 - d. Serial number.
 - e. Range.
 - f. Calibration date.
- 3. Air Handling Unit:
 - a. Location.
 - b. Manufacturer.
 - c. Model.
 - d. Air flow, specified and actual.
 - e. Return air flow, specified and actual.
 - f. Outside air flow, specified and actual.
 - g. Total static pressure (total external), specified and actual.
 - h. Inlet pressure.
 - i. Discharge pressure.
 - j. Fan RPM.
 - k. Cooling Coil Data:
 - 1) Entering air DB temperature, design and actual.
 - 2) Entering air WB temperature, design and actual.
 - 3) Leaving air DB temperature, design and actual.
 - 4) Leaving air WB temperature, design and actual.
 - 5) Water flow, design and actual.
 - 6) Water pressure drop, design and actual.
 - 7) entering water temperature, design and actual
 - 8) Leaving water temperature, design and actual.
 - 9) Air pressure drop, design and actual.
 - l. Heating Coil Data:
 - 1) Water flow, design and actual.
 - 2) Water pressure drop, design and actual.
 - 3) Entering water temperature, design and actual.
 - 4) Leaving water temperature, design and actual.
 - 5) Entering air temperature, design and actual.
 - 6) Leaving air temperature, design and actual.
 - 7) Air pressure drop, design and actual.
 - m. Humidifier Data:
 - 1) Air pressure drop, design and actual.
 - n. Sound Level Data at Operating Condition.
- 4. Exhaust Fan Data:
 - a. Location.
 - b. Manufacturer.
 - c. Model.

- d. Service.
- e. Air flow, specified and actual.
- f. Total static pressure (total external), specified and actual.
- g. Inlet pressure.
- h. Discharge pressure.
- i. Fan RPM.
- j. Sound level data at operating condition.
- 5. Electric Motors:
 - a. Manufacturer.
 - b. HP/BHP.
 - c. Phase, voltage, amperage, nameplate, actual, no load.
 - d. RPM.
 - e. Service factor.
 - f. Starter size, rating, heater elements as applicable.
- 6. V-Belt Drive:
 - a. Identification/location.
 - b. Required driven RPM.
 - c. Driven sheave, diameter and RPM.
 - d. Belt, size and quantity.
 - e. Motor sheave, diameter and RPM.
 - f. Center-to-center distance; maximum, minimum, and actual.
- 7. Duct Traverse:
 - a. System zone/branch air supply.
 - b. Duct size.
 - c. Area.
 - d. Design velocity.
 - e. Design air flow.
 - f. Test velocity.
 - g. Test air flow.
 - h. Duct static pressure.
 - i. Air temperature.
 - j. Air correction factor.
- 8. Air Monitoring Station Data:
 - a. Identification/location.
 - b. System served.
 - c. Size.
 - d. Area.
 - e. Design velocity.
 - f. Design air flow.
 - g. Test velocity.
 - h. Test air flow.
- 9. Air Distribution Test Sheet:
 - a. Air terminal number.
 - b. Room number/location.
 - c. Terminal type.
 - d. Terminal size.
 - e. Area factor.
 - f. Design velocity.

- g. Design air flow.
 - h. Test (final) velocity.
 - i. Test (final) air flow.
 - j. Percent of design air flow.
 - k. Sound data at operating condition.
 - l. Room pressure.
10. Terminal Unit Data:
- a. Manufacturer
 - b. Type (constant, variable)
 - c. Identification/number
 - d. Location.
 - e. Model.
 - f. Size.
 - g. Minimum static pressure.
 - h. Minimum design air flow.
 - i. Maximum design air flow.
 - j. Maximum actual air flow.
 - k. Inlet static pressure.
 - l. Sound data at operating condition.
 - m. Terminal unit controller, setpoints, control variable and monitor I/O.
11. Pump Data:
- a. Type.
 - b. Identification/number.
 - c. Manufacturer.
 - d. Size/model.
 - e. Impeller.
 - f. Service.
 - g. Design flow rate, pressure drop, BHP.
 - h. Actual flow rate, pressure drop, BPH.
 - i. Discharge pressure.
 - j. Suction pressure.
 - k. Total operating head pressure.
 - l. Shutoff, discharge and suction pressures.
 - m. Shutoff, total head pressure.
12. Cooling Tower:
- a. Tower identification/number.
 - b. Manufacturer.
 - c. Model.
 - d. Rated capacity.
 - e. Entering air WB temperature, specified and actual.
 - f. Leaving air WB temperature, specified and actual.
 - g. Ambient air DB temperature.
 - h. Condenser water entering temperature.
 - i. Condenser water leaving temperature.
 - j. Condenser water flow rate.
 - k. Fan RPM.
 - l. Octave band sound power data at operating condition
13. Chillers:
- a. Identification/number.
 - b. Manufacturer.

- c. Capacity.
- d. Model.
- e. Evaporator entering water temperature, design and actual.
- f. Evaporator leaving water temperature, design and actual.
- g. Evaporator pressure drop, design and actual.
- h. Evaporator water flow rate, design and actual.
- i. Condenser entering water temperature, design and actual.
- j. Condenser leaving water temperature, design and actual.
- k. Condenser pressure water temperature, design and actual.
- l. Condenser water flow rate, design and actual.
- 14. Water Flow Measuring Station Data:
 - a. Identification/station.
 - b. Location.
 - c. Size.
 - d. Manufacturer.
 - e. Model.
 - f. Design flow rate.
 - g. Design pressure drop.
 - h. Actual/final pressure drop.
 - i. Actual/final flow rate.
 - j. Station calibrated setting.
- 15. Boiler:
 - a. Boiler manufacturer.
 - b. Model.
 - c. Type.
 - d. Operating pressure.
 - e. leaving water temperature
 - f. Ambient temperature.
 - g. Amp draw
 - h. Entering feedwater temperature.

1.8 PROJECT RECORD DOCUMENTS

- A. Accurately record actual locations of flow measuring stations, balancing valves and setting.

1.9 SEQUENCING AND SCHEDULING

- A. Sequence work to commence after completion of systems and schedule completion of work before Substantial Completion of Project.

1.10 PREINSTALLATION CONFERENCE

- A. Convene a conference 2 weeks prior to commencing Work of this Section, with Construction Manager, electrical, controls and instrumentation, and mechanical equipment installation Contractors.

PART 2 - PRODUCTS

2.1 TESTING AND BALANCING AGENCY

- A. Refer to Quality Assurance, Item 1.10 above.

2.2 INSTRUMENTS AND EQUIPMENT

- A. All instruments and equipment required for tests shall be provided as part of the scope of work. Instrumentation to be used shall be in accordance with the descriptions given in each test procedure. Instrumentation shall have a demonstrated accuracy and sensitivity for the test procedure. Instrumentation shall be calibrated according to the manufacturer's recommendations and so certified and documented at the time of the test. Instrumentation shall include, but not be limited to:
 - 1. Manometer (plus/minus 0.01 inch accuracy).
 - 2. Thermometers (dry bulb and wet bulb).
 - 3. Relative humidity indicators.
 - 4. Velometer (air flow meter).
 - 5. Temperature and humidity recorder.
 - 6. Sound level meter.
 - 7. Flow indicators and pressure gauges.
 - 8. Ammeter and voltmeter.
- B. Provide additional balancing devices as required.

PART 3 - EXECUTION

3.1 PERSONNEL QUALIFICATIONS

- A. A TAB personnel qualification and certification program, meeting the following requirements, shall be maintained by the Test and Balance Contractor.
 - 1. The Test and Balance Contractor shall establish a written practice for the control and administration of TAB personnel training, examination, and certification.
 - 2. The Test and Balance Contractor's written practice shall include a written examination for each personnel TAB level.
- B. There shall be three levels of qualifying TAB personnel for each TAB category hereinafter listed:
 - 1. TAB Level I: A TAB Level I individual shall be a high school graduate and shall have sufficient training and experience to perform the necessary tests in the TAB Category for which he is certified. A TAB Level I

individual shall take test data and shall be responsible to a TAB Level II or TAB Level III Engineer in the applicable TAB Category.

2. TAB Level II:

- a. A TAB Level II individual shall be (1) a graduate of a 4 year accredited engineering college or university, and shall have 2 years of experience in engineering, or equivalent in manufacturing, construction, and installation activities, or (2) be a high school graduate, having 4 years of experience in engineering activities or equivalent. In either case, 2 years of this experience shall be in TAB testing and procedures.
- b. A TAB Level II individual shall direct and carry out tests in the category for which he is certified. He must also be able to set up and calibrate instruments, interpret test results, modify procedures as necessary, analyze data, and record the test results on appropriate forms.

3. TAB Level III:

- a. A TAB Level III individual shall be (1) a graduate of a 4-year accredited engineering or science college or university, plus 5 years of experience in engineering, or equivalent manufacturing, construction, and installation activities, or (2) be a high school graduate, 10 years of experience in engineering, or equivalent manufacturing, construction, and installation activities. In either case, 2 years of this experience shall be in TAB testing and procedures.
- b. A TAB Level III Engineer shall establish test procedures, designate test methods and procedures, and interpret test results. He shall be capable of evaluating the results, not only in terms of existing procedures, but also shall possess sufficient practical background in applicable HVAC technology to assist in establishing test procedures when none are otherwise available. He shall be responsible for conducting examinations of TAB Level I and TAB Level II personnel.

C. TAB personnel shall be qualified and certified in the following generic TAB categories:

- 1. Air Systems and Associated Equipment.
- 2. Hydronic Systems and Associated Equipment.
- 3. Sound.
- 4. Vibration.

D. Certification requirements of all levels of TAB test personnel shall be performed by the Test and Balance Contractor. Certifications shall indicate the category and level of qualification in that category. Written

statements verifying individual qualifications shall be available for review.

3.2 INSTRUMENTS

- A. Instruments used in testing mechanical systems and equipment shall be as recommended by the AABC, AMCA, ASHRAE, or NEBB. Test instruments used shall be initially and periodically checked thereafter to verify their calibration accuracy as described in the AABC or NEBB procedures. Provide calibration verification of each test instrument with each test report.
- B. Test equipment shall be furnished by the Test and Balance Contractor and shall remain his property.

3.3 EXAMINATION

- A. Before commencing work, verify that systems are complete and operable. Start-up per Section 15906 shall be complete prior to commencing work of this Section. Ensure the following.
 - 1. Equipment is operable and in a safe and normal condition.
 - 2. Temperature control systems are installed complete and operable.
 - 3. Proper thermal overload protection is in place for electrical equipment.
 - 4. Final filters are clean and in place. If required, install temporary media in addition to final filters.
 - 5. Duct systems are clean of debris.
 - 6. Correct fan rotation.
 - 7. Fire and volume dampers are in place and open.
 - 8. Coil fins have been cleaned and combed.
 - 9. Access doors are closed and duct end caps are in place.
 - 10. Air outlets are installed and connected.
 - 11. Duct system leakage has been minimized.
 - 12. Hydronic systems have been flushed, filled, and vented.
 - 13. Correct pump rotation.
 - 14. Proper strainer baskets are clean and in place.
 - 15. Service and balance valves are open.
 - 16. All valves to full open position. Set mixing valve, if any, to full system flow.
 - 17. Examine water in system and determine if water has been treated and cleaned.
 - 18. Set all temperature controls so all cooling coils are calling for full cooling.
 - 19. Set all temperature controls so all heating coils and reheating coils are calling for full heating.
 - 20. Check operation of all automatic valves.

- B. Report any defects or deficiencies noted during performance of services to Construction Manager.
- C. Promptly report abnormal conditions in mechanical systems or conditions which prevent system balance.
- D. If, for design reasons, system cannot be properly balanced, report as soon as observed.
- E. Beginning of work means acceptance of existing conditions.

3.4 INSTALLATION TOLERANCES

- A. Adjust air handling systems to plus or minus five percent for supply, return and exhaust systems from figures indicated on the Drawings. Cleanroom air handling systems shall be adjusted to plus 10 percent or minus 0 percent.
- B. Adjust hydronic systems to plus or minus 10 percent of design conditions indicated on the drawings.

3.5 ADJUSTING

- A. Adjust all systems to meet the performance data requirements as scheduled on the drawings and as specified herein.
- B. Recorded data shall represent actually measured or observed condition.
- C. Permanently mark settings of valves, dampers, and other adjustment devices allowing settings to be restored. Set and lock memory stops.
- D. After adjustment, take measurements to verify balance has not been disrupted or that such disruption has been rectified.
- E. Leave systems in proper working order, replacing belt guards, closing access doors, closing doors to electrical switch boxes, and restoring thermostats to specified settings.
- F. At final inspection, recheck random selections of data recorded in report. Recheck points or areas as selected and witnessed by Construction Manager or its designated representative(s).

3.6 GENERAL PROCEDURES

- A. Systems and equipment as listed in the Specifications shall be tested and balanced in accordance with qualified procedures from the AABC or NEBB Standards.

- B. Procedures for each system test and equipment test shall be maintained on file by the Test and Balance Contractor and shall be readily available to the Engineer if requested.
- C. Procedures used in tests shall be included in the submitted report.
- D. Test and Balance Contractor shall coordinate with the controls contractor for setting and/or positioning of control devices pertinent to facilitating testing and adjustment of all mechanical systems.

3.7 PRELIMINARY PROCEDURES

- A. Do all initial planning, including procurement of all available data, study of all systems, instrument selection and calibration, assembling all requisite report forms, study of construction documents and recording pertinent data on report forms, and performing all calculations possible.
- B. Inspect systems and equipment utilizing appropriate standards checklists to assure construction is complete and ready for balancing.
- C. Confirm that equipment installed matches data on report forms, including manufacturer, model, type, size, capacity, motor horsepower, rpm, etc.
- D. Maintain quality control during the execution of work through final completion.

3.8 AIR SYSTEM PROCEDURES

- A. Follow appropriate AABC or NEBB procedures for testing and balancing supply, return and exhaust air systems and systems with economizers and other energy conservation features where included as part of the Project.
- B. Specific systems shall include, but not necessarily be limited to, single duct constant or variable volume systems; pressure independent, variable volume systems; makeup air systems, recirculation air systems, associated return air systems, general, and process exhaust systems.
- C. Conduct tests for central air-handling distribution systems at design air flow with simulated changeout filter resistance.
- D. For variable volume systems, coordinate with equipment supplier to properly set maximum and minimum variable frequency drive speed settings. Maximum speed setting

shall correspond to full load current times motor service factor or maximum airflow at specified fan head, whichever is limiting.

- E. After each air system is balanced, indicate each damper set position with a fully visible, painted red or similar, permanent mark.

3.9 HYDRONIC SYSTEM PROCEDURES

- A. Follow appropriate AABC or NEBB procedures for testing and balancing of individual systems involved.
- B. Central Utility Plant equipment, such as boilers, chillers, compressors, cooling towers, etc., shall be started by and operated under the direct supervision of the Construction Manager.
- C. Specific systems shall include, but not necessarily be limited to, chilled and/or hot water systems, heat exchangers, pumps, and associated appurtenances.
- D. Test and balance equipment provided with manual balancing devices, including air-handling unit heat exchanger coils, terminal reheat coils, fan coil units, and unit heaters. Note that if automatic balancing devices are used, testing and balancing of equipment specifically provided with such devices shall be omitted.
- E. Mark or score balancing devices, gauges, and other adjustable devices at their final set points.

3.10 VIBRATION TESTS

- A. General:
 - 1. The Test and Balance Contractor shall perform vibration tests on all rotating equipment 3 hp and greater. The tested equipment shall comply with the acceptance criteria specified in Section 15240. Non-complying equipment shall be brought to the immediate attention of the Construction Manager.
 - 2. Equipment and systems must be tested, adjusted and balanced prior to vibration testing of the equipment in that system.
- B. Instruments: Reed vibrometers, vibrometers, vibration meters, vibrographs, and vibration analyzers are the various types of instruments available for measuring frequency and/or amplitude. In addition to the conventional instruments (stethoscopes, tachometers, and stroboscopes) the Test and Balance Contractor shall have, as a minimum, an instrument which measures both amplitude and frequency. The choice of instrumentation is a

function of the type of testing to be performed. Specific instrumentation used in a test shall be subject to review by the Engineer.

- C. Procedures: The following steps should be followed to assure that equipment vibration is properly tested:
 - 1. Determine operating speeds of equipment from nameplates, Drawings, or speed measuring devices such as tachometers or stroboscopes and indicate on test form. For any equipment where driving speed (motor) is different from driven speed (fan wheel, rotor, impeller) because of belt driver or gear reducers, indicate both driving and driven speeds.
 - 2. Assure that vibration isolation system is functioning properly.
 - 3. Operate equipment and make visual and audible check for an apparent rough operation. Check all bearings with stethoscope. Correct any condition of defective bearings, misalignment, or obvious rough operation before proceeding further. If not corrected, equipment may be considered unacceptable.
 - 4. Make vibration measurements in accordance with the method specified in Section 15240.
 - 5. Evaluate measurements in accordance with Section 15240. Measurement results must incorporate corrections for inertial mass as specified in Section 15240.
 - 6. Equipment not complying with the most stringent of the criteria in 15240 or ASHRAE vibration tolerances shall be corrected at the manufacturer's expense. Equipment shall then be re-tested and measurement results reported.

3.11 REPORT FORMS

- A. Report forms, as they relate to the specific Project, shall be fully implemented, including a sealed certification. Report forms shall include, but not be limited to: coils, fans, duct systems, air outlets, terminal units, rooftop equipment, heat exchangers, pumps, and instrument calibration.
- B. Report forms shall be standard forms as published by AABC or NEBB.

3.12 RE-TESTING

- A. During the period following completion of a specific test, and within the Project time frame, the Engineer may, at his discretion, request a recheck or resetting of any outlet, supply air fan, or exhaust fan as originally covered in the Scope of Work of this Specification. Re-testing procedures, personnel qualifications, and report

requirements shall be the same as the original requirements.

- B. Testing, adjusting and balancing not done in accordance with approved procedures, or items found not to be in conformance to the drawings or specifications, shall be corrected and re-tested without additional cost to Construction Manager.

- END -